

Roll No.

TEE-803

**B. TECH. (EE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024**

ADVANCED ELECTRIC DRIVES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the working of Half-Bridge PWM inverter with suitable diagram. (CO1, CO2)
- (b) Discuss the Constructional feature of Brush less dc Motor in details. (CO1, CO2)
- (c) What do you understand by Single Phase Full-Bridge Inverter discuss its application ? (CO1, CO2)
2. (a) Discuss the merits and demerits of BLDC motor. (CO3, CO1)
- (b) Discuss the unipolar switching technique used for Sinusoidal-Pulse Width Modulation. (CO3, CO1)

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- (c) Discuss the working of Brush Less Permanent Magnet Square wave motor. (CO3, CO1)
- 3. (a) Explain the working of Bipolar and Modified Bipolar PWM Schemes with Zero Sequence Voltage. (CO3, CO5)
- (b) Discuss the role of Demagnetization curve and Electromechanical transfer. (CO3, CO5)
- (c) Explain working of dc/dc boost chopper with suitable diagram. (CO3, CO5)
- 4. (a) Explain the working of half bridge Voltage Source Inverters. (CO1, CO6)
- (b) Obtain the equation for open circuit emf of a synchronous reluctance motor. (CO1, CO6)
- (c) Enumerate the advantages and disadvantages of synchronous reluctance motor. (CO1, CO6)
- 5. (a) State and explain the working of synchronous reluctance motor. (CO4, CO6)
- (b) Deduce the mathematical relation for Low and High Magnetic Reluctances in Magnetic Circuits and discuss their influences. (CO4, CO6)
- (c) Draw the phasor diagram of synchronous reluctance motor and hence deduce the relation for torque produced. (CO4, CO6)

END SEMESTER EXAMINATION 2024

Name of the Program: B.Tech. (EE).

Semester: 8th

Name of the Course: Expert System and Fuzzy Logic

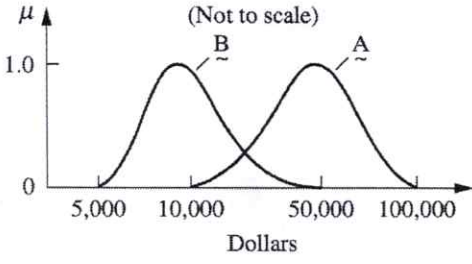
Course Code: TOE 801

Maximum Marks: 100

Time: 3 hrs

Note

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

1	20 Marks	
(a)	Explain the various types of membership function used to represent fuzzy set. Also show the mathematical representation of these membership function.	CO2 CO3
(b)	In determining corporate profitability, many construction companies must make decisions based upon the particular client's spending habits, such as the amount the client spends and their capacity for spending. Many of these attributes are fuzzy. A client which spends a "large amount" is considered to be "profitable" to the construction company. A "large" amount of spending is a fuzzy variable, as is a "profitable" return. These two fuzzy sets should have some overlap, but they should not be defined on an identical range. $\tilde{A} = \{\text{"large" spenders}\}$ $\tilde{B} = \{\text{"profitable" clients}\}$ For the two fuzzy sets shown in Fig. P2.5, find the following properties graphically:  i. all clients deemed profitable or who are large spenders. ii. all clients deemed profitable and large spenders. iii. those clients (i) deemed not profitable, and (ii) deemed not large spenders (separately). iv. entities deemed profitable clients, but not large spenders. v. $\tilde{A} \cup \tilde{B} = \tilde{A} \cap \tilde{B}$ (De Morgan's principle).	CO2, CO3, CO4, CO5
(c)	The membership function distribution of a fuzzy set is assumed to follow a Gaussian distribution with mean, $m = 100$ and standard deviation of 20. Determine 0.6-cut of this distribution.	CO2, CO4 CO 5
2	20 Marks	
(a)	Define the following features of the membership function. i. Core of a membership function ii. Support of a membership function iii. Boundaries of a membership function iv. Normal fuzzy set v. Convex fuzzy set vi. Crossover points of a membership function vii. Height of a fuzzy set	CO2

(b)	Many products, such as tar, petroleum jelly, and petroleum, are extracted from crude oil. In a newly drilled oil well, three sets of oil samples are taken and tested for their viscosity. The results are given in the form of the three fuzzy sets B_1, B_2, and B_3, all defined on a universe of normalized viscosity, as shown in Figs. 2(a), 2(b) and 2(c). find the most nearly representative viscosity value for all three oil samples, and hence find z^* for the three fuzzy viscosity sets using the centroid method. FIGURE 2 (a) Membership in viscosity of oil sample 1, B_1. FIGURE 2 (b) Membership in viscosity of oil sample 2, B_2. FIGURE 2 (c) Membership in viscosity of oil sample 3, B_3. find z^* using the centroid method and the Center of sums method.	CO2 CO5
(c)	with a velocity v (meters per second), then the kinetic energy k (in joules) is given by the equation $k = \frac{1}{2}mv^2$. Suppose we model the mass and velocity as inputs to a system (moving body) and the energy as output, then observe the system for a while and deduce the following two disjunctive rules of inference based on our observations: Rule 1 : IF x_1 is $\sim A_1$ (small mass) and x_2 is $\sim A_{12}$ (high velocity), THEN y is $\sim B_1$ (medium energy). Rule 2 : IF x_1 is $\sim A_2$ (large mass) or x_2 is $\sim A_2$ (medium velocity), THEN y is $\sim B_2$ (high energy). Fire the above two rule to give one fuzzy output. Assume the membership function for A_1, A_{21}, A_{12}, A_2, B_1, and B_2	CO2

3	20 Marks	
(a)	Explain different defuzzification methods used in fuzzy logic. i. <i>Max membership principle:</i> ii. <i>Centroid method:</i> iii. <i>Weighted average method:</i> iv. <i>Mean max membership:</i>	CO1
(b)	Draw the block diagram of fuzzy logic controller. Explain the function of each component. Also give a suitable example.	CO1
(c)	For the following navigation problem design a Mamdani fuzzy logic controller. Let us consider a problem scenario related to navigation of a mobile robot in presence of four moving obstacle. Obstacle 2 is found to be the most critical one. The aim is to develop a fuzzy logic-based motion planner that will be able to generate the collision free path for the robot. There are two inputs, namely the distance between the robot and the obstacle (D), and the angle ($\angle GSO_2$) for the motion planner and it will generate one output, that is, deviation. Distance is represented using four linguistic terms, namely, very near (VN), Near (NR), far (FR), and very far (VFR). Whereas the input (i.e., angle) and the output (i.e., deviation) are expressed with the help of five linguistic terms such as Ahead (A), Ahead left (AL), Left (LT), Ahead right (AR) and Right (RT). Determine the output if distance $D = 1.04$ m and angle ($\angle GSO_2$) = 30, Using Mamdani approach. Use any one of the defuzzification method.	CO1, CO5
4	20 Marks	
(a)	What is an Expert-systems? what are the characteristics/attributes of Expert systems?	CO1
(b)	Draw the general structure of an expert system. Also explain its components.	CO4
(c)	What is database and rule base in fuzzy logic system. Explain the fuzzy reasoning process with a suitable example	CO 4
5	20 Marks	
(a)	What is knowledge elicitation. Explain the different knowledge elicitation techniques in expert system.	CO1
(b)	Explain the Techniques for Representing the Expert Knowledge in expert system.	CO2
(c)	What is an expert system shell? Explain the following. MYCIN EMYCIN ROSIE	CO2,

